

# Figure 3 — Weighted Gene Co-Expression Network Analysis (WGCNA)

Topological module identification and trait correlations in real *Mycobacterium marinum* OSD-528

## A — Empirical Co-Expression Modules

- MEturquoise: Microgravity-Associated Core** ( $n = 35$ )  
Key hubs: MMAR\_RS12120 ( $k = 35.3$ ), RS21560 ( $k = 35.0$ ),  
MMAR\_RS12120 ( $k = 35.3$ ), RS21560 ( $k = 35.0$ )
- MEblue: Kinematic Simulator Disparity** ( $n = 5$ )  
Key hubs: MMAR\_RS29685 ( $k = 15.4$ ), RS02330 ( $k = 14.8$ ),  
MMAR\_RS29685 ( $k = 15.4$ ), RS02330 ( $k = 14.8$ )
- MEbrown: Secretion & Oxidoreductases** ( $n = 6$ )  
Key hubs: MMAR\_RS16730 ( $k = 10.8$ ), RS23800 ( $k = 10.6$ ),  
MMAR\_RS16730 ( $k = 10.8$ ), RS23800 ( $k = 10.6$ )
- MEyellow: Stress Dormancy & Transcriptional** ( $n = 3$ )  
Key hubs: MMAR\_RS13930 ( $k = 6.3$ ), RS09685 ( $k = 6.2$ ), RS  
MMAR\_RS13930 ( $k = 6.3$ ), RS09685 ( $k = 6.2$ )
- MEgreen: Rotational Shear Adaptation** ( $n = 3$ )  
Key hubs: MMAR\_RS11565 ( $k = 11.9$ ), RS11250 ( $k = 11.8$ ),  
MMAR\_RS11565 ( $k = 11.9$ ), RS11250 ( $k = 11.8$ )

## B — Empirical Module-Trait Correlations

|             | Microgravity | Cyностat vs RPM |
|-------------|--------------|-----------------|
| MEturquoise | $r = -0.77$  | $r = 0.21$      |
| MEblue      | $r = -0.26$  | $r = 0.93$      |
| MEbrown     | $r = 0.03$   | $r = -0.15$     |
| MEyellow    | $r = -0.40$  | $r = -0.03$     |
| MEgreen     | $r = 0.05$   | $r = 0.55$      |